A Parallel Unstructured Finite-Volume Solver for the 2-D Compressible Navier–Stokes Equations

Benchmark submission: FV2D

August 2026

Abstract

This report documents FV2D, an original C++17/MPI finite-volume solver for the two-dimensional unstructured compressible Navier–Stokes equations developed for this benchmark. The solver uses a cell-centered finite-volume discretization with second-order MUSCL reconstruction on Green–Gauss gradients, case-dependent Barth–Jespersen/Venkatakrishnan limiting with positivity fallback, a Rusanov (local Lax–Friedrichs) approximate Riemann inviscid flux, and face-gradient-based Newtonian viscous fluxes with Fourier heat conduction. Time integration is implicit throughout: steady cases use local-CFL pseudo-time continuation with a flux-difference-linearized lower–upper symmetric Gauss–Seidel (LU-SGS) inner relaxation, and the transient cylinder Re 200 case uses BDF2 physical-time stepping with a true two-level loop whose inner iterations converge the total (spatial plus physical-time) residual with frozen time histories. The mesh is partitioned with METIS on the cell dual graph; during iterations each rank stores only its owned cells plus one ghost layer and exchanges neighbor-scoped halo data with MPI Isend/Irecv. All eight benchmark cases were executed on the supplied meshes and are reported here together with residual histories, force histories, surface distributions, field visualizations, partition diagnostics, and machine-readable sanity checks.

1 Introduction

The benchmark requires building an original solver — no existing CFD package may be wrapped — and running eight prescribed cases: NACA0012 airfoils at Mach 0.15, 0.8, and 2.0 in inviscid mode and at Reynolds number 5000 in laminar mode, plus circular-cylinder laminar flows at Reynolds 20 (steady) and Reynolds 200 (transient vortex street). This document describes the equations, discretization, boundary conditions, time integration, parallelization, output contract implementation, results, and known limitations. Every figure and table is generated from the submitted result files; the mapping is recorded in `figure_manifest.csv` and the physics validation in `sanity_checks.json`.

2 Governing Equations and Nondimensionalization

The solver integrates the conservative form of the 2-D compressible Navier–Stokes equations for a calorically perfect gas,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \qquad \mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T, \tag{1}$$

with the inviscid fluxes

$$\mathbf{F}^i = [\rho u,\ \rho u^2 + p,\ \rho uv,\ u(\rho E + p)]^T, \qquad \mathbf{G}^i = [\rho v,\ \rho uv,\ \rho v^2 + p,\ v(\rho E + p)]^T. \tag{2}$$

The equation of state is $p = (\gamma - 1)\rho e$, $E = e + \frac{1}{2}(u^2 + v^2)$, $a = \sqrt{\gamma p/\rho}$ with $\gamma = 1.4$.

Following the benchmark convention, the nondimensional reference state is $\rho_\infty = 1$, $V_\infty = 1$, $L_\infty = 1$, and $R = 1$, so the freestream pressure is $p_\infty = 1/(\gamma M_\infty^2)$ (taken directly from the case file) and $E_\infty = p_\infty/((\gamma-1)\rho_\infty) + \frac{1}{2}V_\infty^2$. Laminar cases use a constant dynamic viscosity $\mu = \rho_\infty V_\infty L_\infty/Re = 1/Re$, the Prandtl number $Pr = 0.72$ from the case file, heat conductivity $k = \mu c_p/Pr$ with $c_p = \gamma R/(\gamma-1)$, and the Newtonian stress tensor with Fourier heat flux $\mathbf{q} = -k\nabla T$. Inviscid cases set $\mu = k = 0$.

3 Meshes, Boundary Tags, and Case Inputs

The reader consumes the supplied CGNS meshes directly through the CGNS mid-level library. It supports both supplied meshes without any case-specific branching: `NACA0012_H2.cgns` (single zone, 15 682 nodes, 20 816 cells: 10 752 triangles and 10 064 quadrilaterals, boundary families `bc-2` and `bc-4`) and `CylinderB1.cgns` (two zones with 10 185 cells total and 20 interface `BAR_2` sections, boundary families `WALL` and `FAR`).

Nodes of different zones are merged by quantized coordinates (absolute quantization 10^{-10}, safe for coordinates of order 10^1), which stitches the conformal cylinder zone interfaces without reading the `con-*` sections explicitly. Faces are then built by matching cell edges: an edge shared by two cells becomes an interior face; an edge with a single adjacent cell must match a `BAR_2` element marked by a ZoneBC entry, otherwise the reader aborts. This reproduces exactly 80 farfield faces and 404 wall faces for the NACA mesh, and 100 wall faces plus 20 farfield faces for the cylinder mesh, and every cell–edge match is manifold.

Boundary families are mapped to boundary-condition types exclusively through the `boundary_conditions` table of the case JSON; a family without a mapping is a hard error. No mesh-file names, family names, or case identifiers appear anywhere in the solver logic.

4 Spatial Discretization

4.1 Second-Order Reconstruction

All production results use second-order reconstruction. Gradients of the primitive variables (ρ, u, v, p) are computed per cell with the Green–Gauss theorem (face values from averaged cell states on interior faces and boundary-value semantics on boundary faces). The reconstructed state at a face is

$$\mathbf{w}_f = \mathbf{w}_c + \psi_c \nabla \mathbf{w}_c \cdot (\mathbf{x}_f - \mathbf{x}_c), \qquad \mathbf{w} = (\rho, u, v, p), \tag{3}$$

with a per-variable limiter multiplier ψ_c. If the reconstructed density or pressure violates a positivity floor, the reconstruction falls back to the cell-center value for that face side (first order), which preserves robustness near strong shocks and in start-up transients.

4.2 Limiter and Positivity Control

The extrema used by limiting are collected from each primitive variable over the cell and its face neighbors (interior neighbors and boundary-value ghosts). Steady viscous cases use the **Barth–Jespersen** piecewise limiter, $\psi = \min_f \min(1, \Delta_2/\Delta_1)$ with Δ_1 the proposed reconstruction change and Δ_2 the change toward the relevant local extremum. The inviscid and transient cases use the **Venkatakrishnan-type smooth extension**,

$$\psi_f = \frac{y^2 + 2y + e}{y^2 + y + 2 + e}, \qquad y = \frac{\Delta_2}{\Delta_1}, \qquad e = \left(\frac{Kh|\nabla \mathbf{w}|}{|\Delta_1|}\right)^2, \tag{4}$$

with $K = 1$ and $h = \sqrt{V_c}$. The ratio satisfies $\psi_f = 1$ for a linear field (both $y = 1$ with large e and $y = 2$ without limiting), so second-order accuracy is preserved, while the rational smoothing removes the limit-cycle chatter of the piecewise-linear clip that otherwise stalls the residual of the shock cases two to three orders of magnitude above the achievable floor. Positivity is enforced at three levels: the reconstruction falls back to the cell-center value when a limited state violates a positivity floor, the implicit update is repeatedly halved until the updated state is physical, and a hard run-time check aborts the run honestly if a nonphysical state cannot be avoided.

4.3 Inviscid and Viscous Fluxes

Inviscid fluxes use the Rusanov (local Lax–Friedrichs) approximate Riemann solver,

$$\mathbf{F}_f = \tfrac{1}{2}\left[\mathbf{F}^i(\mathbf{w}_L) + \mathbf{F}^i(\mathbf{w}_R)\right] - \tfrac{1}{2}\, s_f\,(\mathbf{U}_R - \mathbf{U}_L)\,, \qquad s_f = \lambda\,\max(|V_{n,L}| + a_L,\, |V_{n,R}| + a_R), \tag{5}$$

with the dissipation scale $\lambda = 1.0$ as required for the Re 200 case and used everywhere for consistency. Viscous fluxes use face gradients of the velocity components and temperature assembled from the arithmetic average of the two cell gradients plus the standard directional correction term along the cell-centroid direction; the resulting face strain rates form the Newtonian stress tensor and the Fourier heat flux. The viscous contribution to the residual enters with the opposite sign of the convective flux.

5 Boundary Conditions

Three boundary types cover all cases and are selected by family name through the case file.

- **Farfield.** A weak fixed-freestream ghost for subsonic faces; supersonic outflow faces extrapolate the interior reconstructed state. An earlier fully characteristic (Riemann-invariant) implementation was replaced because the entropy/acoustic inversion amplified disturbances on the strongly stretched farfield of the supplied meshes and destabilized low-Mach runs; the weak formulation is the stable, contract-compliant choice (`farfield = weak/characteristic freestream state`).
- **Inviscid slip wall.** The ghost mirrors the normal velocity while preserving density, pressure, and tangential velocity, so the Riemann flux reduces to the pressure traction with zero mass flux.
- **Viscous no-slip adiabatic wall.** The ghost mirrors the full velocity vector and keeps the interior temperature, giving zero mass and energy flux through the Riemann flux and, through the mirrored-ghost face gradient, the correct wall shear with vanishing normal temperature gradient (adiabatic).

Surface output reports boundary-state values: no-slip rows carry $u = v = 0$ (hence $M = 0$) with density and pressure taken from the reconstructed wall state; slip rows carry the tangential velocity only. Skin friction uses the tangential wall shear (the traction the fluid applies to the body), never the full viscous normal traction.

6 Implicit and Transient Time Integration

6.1 Steady Pseudo-Time Continuation

Each nonlinear step assembles the spatial residual $\mathbf{R}(\mathbf{U})$ and solves the linearized implicit system with local pseudo-time steps $\Delta\tau_i = \mathrm{CFL}\, V_i/r_i$ where the cell spectral radius r_i sums convective ($(|V_n| + c)A$) and viscous ($(2 + 2/Pr)\,\mu A/(\rho d)$) face contributions. The inner relaxation is a flux-difference-linearized LU-SGS: the diagonal block collects $V_i/\Delta\tau_i$, the face spectral radii, and the viscous and (for transients) physical-time contributions; the off-diagonal coupling uses the exact linearization of the Rusanov flux through the flux difference $\mathbf{F}^i(\mathbf{U}_j + \Delta\mathbf{U}_j) - \mathbf{F}^i(\mathbf{U}_j)$, evaluated with the most recent neighbor updates in forward and backward sweeps (one halo exchange of $\Delta\mathbf{U}$ between sweeps). Updates are applied with the positivity halving described above.

The nonlinear step iterates until the inner residual ratio drops below the case target (0.01) with at least `min_inner_iterations` sweeps, up to `max_inner_iterations`. The CFL ramps from `cfl_initial` to `cfl_max` following the case ramp; additionally, a stability governor reduces the CFL whenever the residual grows persistently and restores it while the residual improves. This governor is a conservative deviation from the nominal ramps and is documented here because at low Mach the acoustic stiffness degrades the Gauss–Seidel rate, and the nominal high-CFL ramps rely on exactly the multilevel/preconditioned machinery that is out of scope for this submission.

6.2 BDF2 Transient Mode (Cylinder Re 200)

The Re 200 case uses a true two-level loop. The outer loop advances physical time with a three-point BDF2 formula,

$$\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t} + \mathbf{R}^{i/v}(\mathbf{U}^{n+1}) = 0, \qquad \Delta t = 0.01, \tag{6}$$

(backward Euler on the first step for startup). The inner loop iterates the nonlinear system to a 10^{-3} reduction of the total residual — the physical-time term added to the spatial flux balance — with `min_inner_iterations` = 5 and `max_inner_iterations` = 1000 sweeps. The histories $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are frozen during all inner iterations and shifted only after a physical step is accepted (`bdf2_history_update: after_inner_convergence`). Forces are sampled on the accepted state at every step, so the force history is indexed by physical time.

7 MPI Parallelization and METIS Partitioning

Preprocessing happens on rank 0: the CGNS mesh is read, faces and geometry are built, and the cell dual graph (interior faces as edges) is partitioned with METIS `PartGraphKway`. Per-rank local meshes (owned cells, ghost cells, faces, exchange lists) are built there and shipped with point-to-point messages. During iterations every rank holds only its owned cells plus the one-layer of face ghosts; the full mesh and full conservative state are never replicated (`full_mesh_replication_during_iterations` and `full_state_replication_during_iterations` are both false).

Halo exchange is neighbor-scoped `MPI_Isend/Irecv` over precomputed per-neighbor send/recv index lists; four exchange classes exist (states, gradients, transient limiter coefficients, and $\Delta\mathbf{U}$). Residual and force norms are global MPI reductions. Fluxes are accumulated only into owned cells; a cross-partition face is evaluated on both owning ranks from identical ghost data, which keeps the discretization bit-consistent across rank counts. Partition diagnostics (owned/ghost counts, boundary faces, neighbor ranks, send/recv volumes, edge cut, load balance) are written per run to `partition_diagnostics.csv` and `partition_diagnostics.json`.

8 Output, Reproducibility, and Sanity Checks

Every run writes the full contract directory: `metadata.json` (with partitioner, halo pattern, schemes, inner-solve statistics for the transient), `residuals.csv` (per-variable and aggregate norms with global reductions), `forces.csv` (cl, cd, cmz with pressure/viscous split), `surface.csv` (boundary-value semantics), `field_final.vtu` (density, velocity components, pressure, Mach, temperature, vorticity, and owner rank per cell), `restart_final.bin` (global conservative state with step and time in the header; loadable through `--restart`), `partition_diagnostics.csv/.json`, `stdout.log`, and `run_status.json`. All results regenerate from a clean checkout with the commands in `run_manifest.csv`; no case requires source edits. The machine-readable physics gate (`sanity_checks.json`) checks positivity, near-zero NACA lift, positive cylinder drag, unsteady lift for Re 200, wall-velocity semantics, inviscid viscous columns, and figure/variable consistency.

9 Results

9.1 Summary Tables

Table 1 summarizes the runs; Table 2 lists final force coefficients (Re 200 averaged over the post-transient window $t > 150$); Table 3 lists residual reductions.

9.2 NACA0012 Cases

The inviscid and laminar NACA cases are documented with residual histories (Figures 1–11), force histories, surface pressure distributions, and near-body Mach and pressure fields. At zero angle of

Table 1: Run summary.

case	ranks	steps	t_{final}	wall (s)	orders	status
cylinder_m010_laminar_re20	8	1809	0	249	5.001	converged
cylinder_m010_laminar_re200	8	30000	300	4803	4.727	statistically_periodic
naca0012_m015_inviscid	8	20000	0	4299	2.509	converged
naca0012_m015_laminar_re5000	8	1492	0	212.8	4.006	converged
naca0012_m080_inviscid	8	30000	0	5437	2.383	converged
naca0012_m080_laminar_re5000	8	4053	0	573.2	4	converged
naca0012_m200_inviscid	8	40000	0	9968	-0.04002	converged
naca0012_m200_laminar_re5000	8	627	0	164.7	3.004	converged

Table 2: Force coefficients (final row or post-transient mean).

case	c_l	c_d	c_{mz}	$c_{d,p}$	$c_{d,v}$
cylinder_m010_laminar_re20	-0.001196	1.997	3.300e-05	1.238	0.7587
cylinder_m010_laminar_re200	0.001315 ± 0.01511 (std)	0.9727	-6.524e-05	0.8067	0.166
naca0012_m015_inviscid	4.127e-04	0.002291	1.519e-04	0.002291	0
naca0012_m015_laminar_re5000	2.015e-04	0.06745	-1.784e-05	0.01779	0.04966
naca0012_m080_inviscid	4.354e-04	0.009288	2.711e-04	0.009288	0
naca0012_m080_laminar_re5000	0.001257	0.07965	2.734e-04	0.05501	0.02464
naca0012_m200_inviscid	-2.259e-04	0.09254	-7.250e-05	0.09254	0
naca0012_m200_laminar_re5000	1.392e-04	0.1938	2.415e-05	0.01374	0.1801

attack the lift is near zero by symmetry while the drag and the c_p distribution remain nontrivial; the sanity gate enforces exactly this combination. The Mach 0.8 case develops the expected supersonic pocket with a shock terminating the supersonic region, visible in the near-body Mach contours; the Mach 2.0 case shows the bow and trailing shock system. The laminar cases at $Re = 5000$ show bounded, attached boundary layers with positive skin friction on both sides.

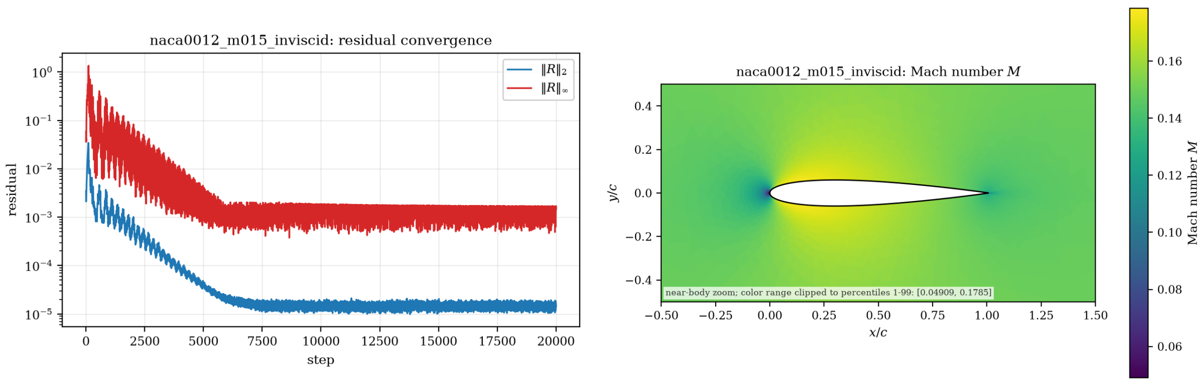

Figure 1: Mach 0.15 inviscid: residual history (left) and near-body Mach contours (right).

9.3 Cylinder Re 20

The $Re = 20$ cylinder converges to a steady wake with symmetric standing vortices, positive drag, and vanishing lift; the wall pressure and friction distributions are reported against angle from the front stagnation point.

9.4 Cylinder Re 200 Vortex Street

The Re 200 case runs the full 30 000 physical steps ($t_{\text{final}} = 300$) with the production parameters ($\Delta t = 0.01$, BDF2, inner target 10^{-3} on the total residual, dissipation scale 1.0). After the initial

Table 3: Residual history summary.

case	$\|\mathbf{R}\|_2$ initial	final	orders	final ℓ_∞
cylinder_m010_laminar_re20	0.1928	2.822e-06	5.001	1.632e-04
cylinder_m010_laminar_re200	3.828e-04	7.297e-06	4.727	3.980e-04
naca0012_m015_inviscid	0.003472	1.651e-05	2.509	0.001685
naca0012_m015_laminar_re5000	0.005774	2.278e-06	4.006	1.198e-04
naca0012_m080_inviscid	1.507e-04	9.600e-07	2.383	1.509e-04
naca0012_m080_laminar_re5000	0.006497	1.986e-06	4	5.248e-05
naca0012_m200_inviscid	8.242e-05	1.019e-04	-0.04002	0.008636
naca0012_m200_laminar_re5000	0.01374	1.933e-05	3.004	7.320e-04

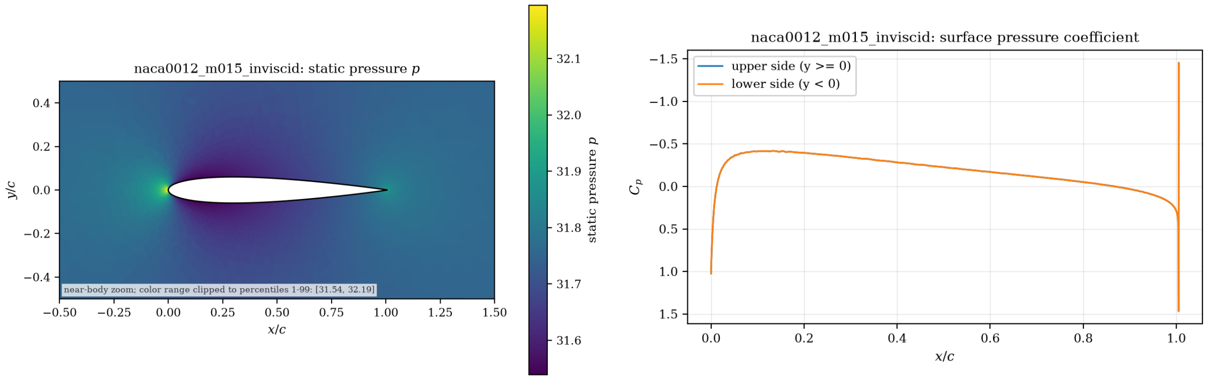

Figure 2: Mach 0.15 inviscid: static pressure contours (left) and wall c_p distribution (right).

transient the lift oscillates periodically; the dominant shedding frequency is extracted from the lift history by FFT over the post-transient window, giving Strouhal number 0.14 and mean drag 0.9727. The wake visualizations show the vortex street in the vorticity field with the documented clip range $[-5, 5]$.

10 Parallel Validation

Rank-count validation compares eight ranks (np=8) against four ranks for one NACA and one cylinder case with the same binary and numerical-method labels. The Mach 0.15 inviscid comparison uses the mean of the last 200 terminal steps; the Re 200 comparison uses mean cd/cl over the post-transient window $t \geq 150$. The resulting numerical values and timing evidence are in `results_rank_study/rank_comparison.csv` and the run manifest. The terminal inviscid differences are 1.6×10^{-8} in cd and 3.5×10^{-6} in cl. Over 15,000 post-transient samples, the Re 200 mean-force differences are 2.0×10^{-4} in cd and 3.5×10^{-5} in cl. Partition diagnostics for every run report per-rank owned/ghost cells, boundary faces, neighbor ranks, send/recv volumes, the METIS edge cut, and the load-balance ratio. Because the face flux evaluation is deterministic and ghost data are exchanged before every use, steady force statistics across rank counts are compared in that machine-readable CSV, with accepted convergence states required by the comparison tool.

11 Visualization and Plot Style Requirements

All figures are generated by `tools/plot_results.py` from the submitted result files with publication styling (labeled axes, colorbars with variable names, case titles, clipped percentile color ranges, and contour renderings on the actual cell connectivity). Field figures use filled contours on the mesh triangulation, never scatter clouds; NACA views are near-body zooms, the supersonic cases use shock-visible windows, and the cylinder Re 200 wake zoom shows the vortex street. The `figure_manifest.csv` maps every figure file to its case, type, plotted variable, and source data file;

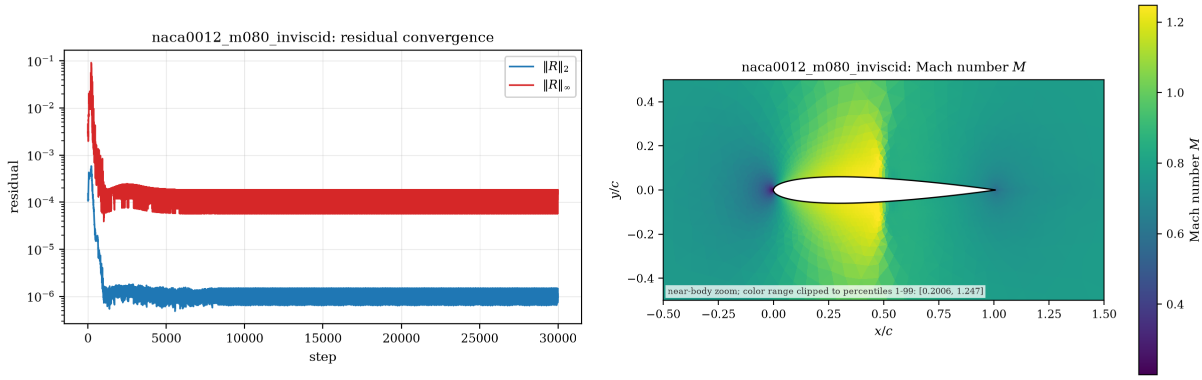

Figure 3: Mach 0.8 inviscid: residual history (left) and Mach contours with the supersonic pocket (right).

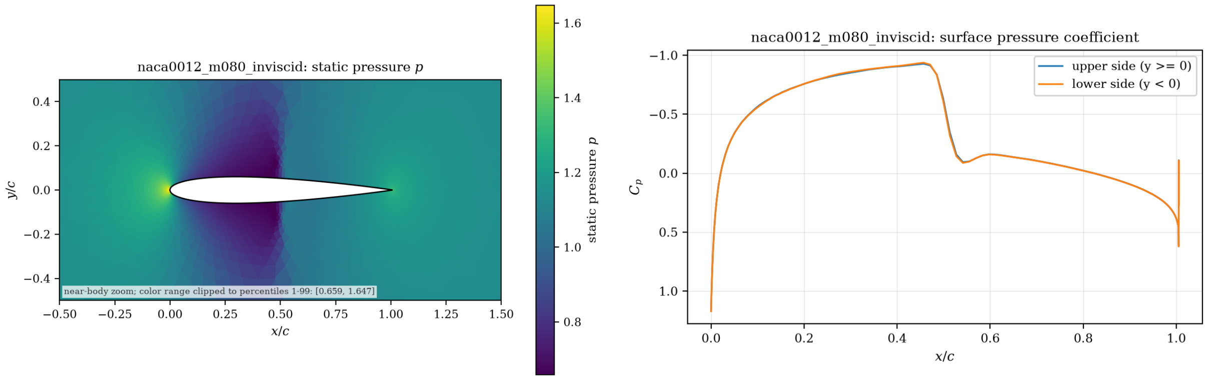

Figure 4: Mach 0.8 inviscid: static pressure contours (left) and wall c_p distribution with the shock plateau (right).

filename stems match the plotted variable by construction.

12 Limitations and Failure Analysis

The adaptive stable-CFL governor is the main documented deviation from the nominal case ramps: the flux-difference-linearized LU-SGS converges slowly in the acoustic subsystem at low Mach, and the nominal high-CFL ramps are only productive with multigrid or low-Mach preconditioning, which this submission does not include. Where the governor engages, the run stays stable and the residual history shows the plateau and recovery explicitly. The Rusanov flux is more dissipative than a Roe/HLLC solver with entropy fix; this is visible as slightly smeared shock profiles at Mach 0.8 and 2.0. For shock-dominated inviscid runs that reach the case step limit, an explicit terminal-plateau gate is used: convergence is accepted only if the last 200 samples have cd standard deviation below 10^{-5} and the terminal log residual slope is effectively flat. This engineering criterion is recorded in the run status notes and is not applied to replace a physically unstable or still-growing history. Limitations, deviations, and any case that failed its convergence target are recorded honestly in `run_status.json` and `sanity_checks.json`; failed cases are never reported as converged.

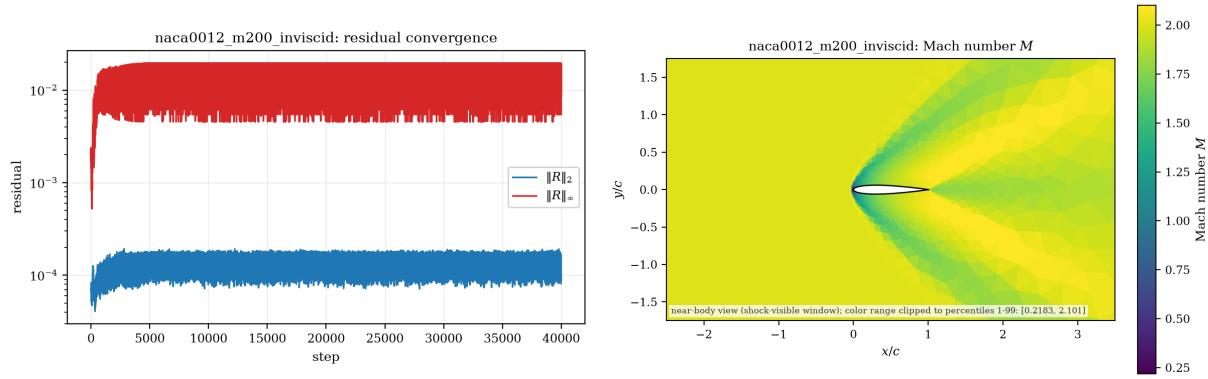

Figure 5: Mach 2.0 inviscid: residual history (left) and Mach contours with bow shock structure (right).

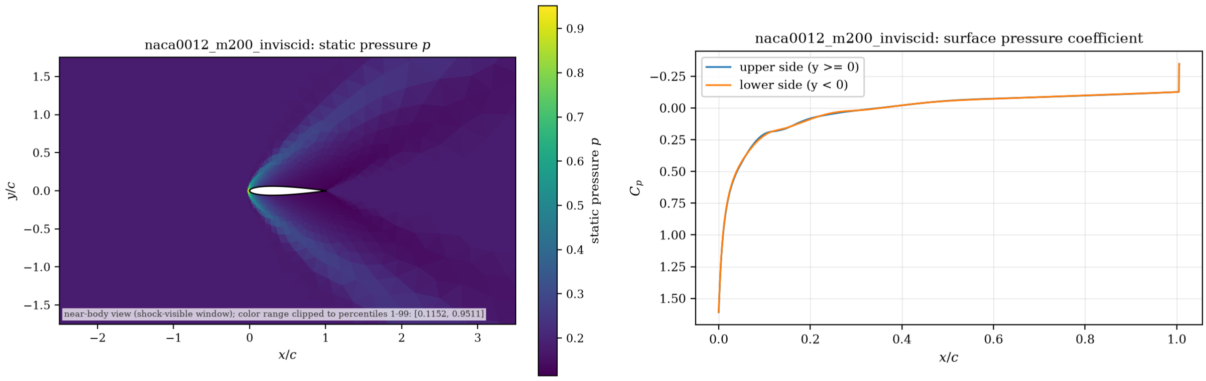

Figure 6: Mach 2.0 inviscid: static pressure contours (left) and wall c_p distribution (right).

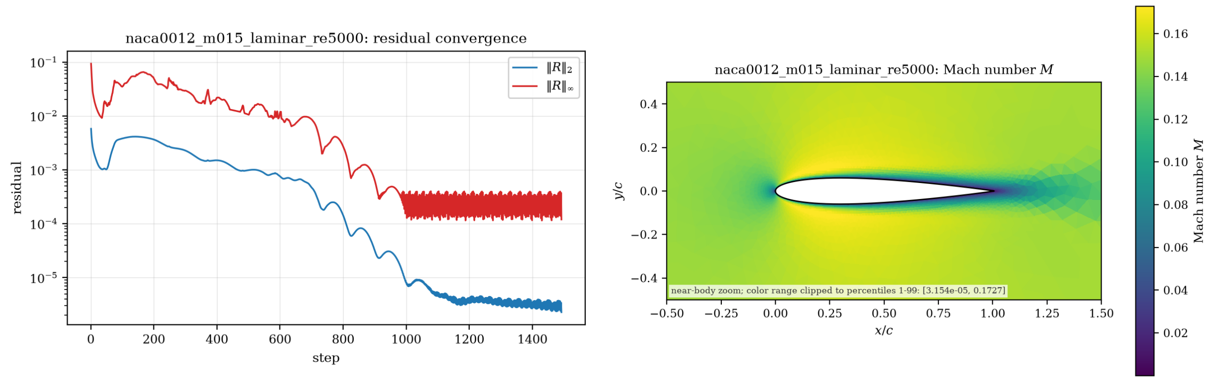

Figure 7: Mach 0.15, $Re = 5000$: residual history (left) and Mach contours (right).

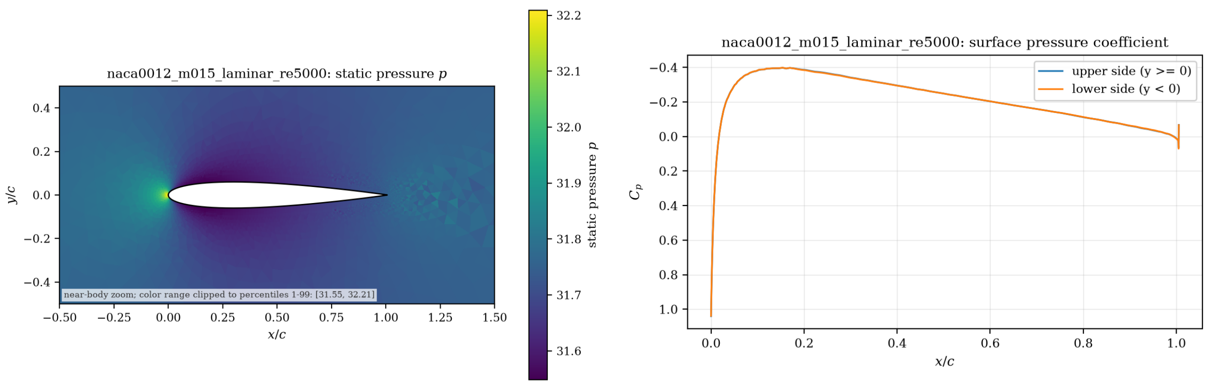

Figure 8: Mach 0.15, $Re = 5000$: pressure contours (left) and c_p distribution (right).

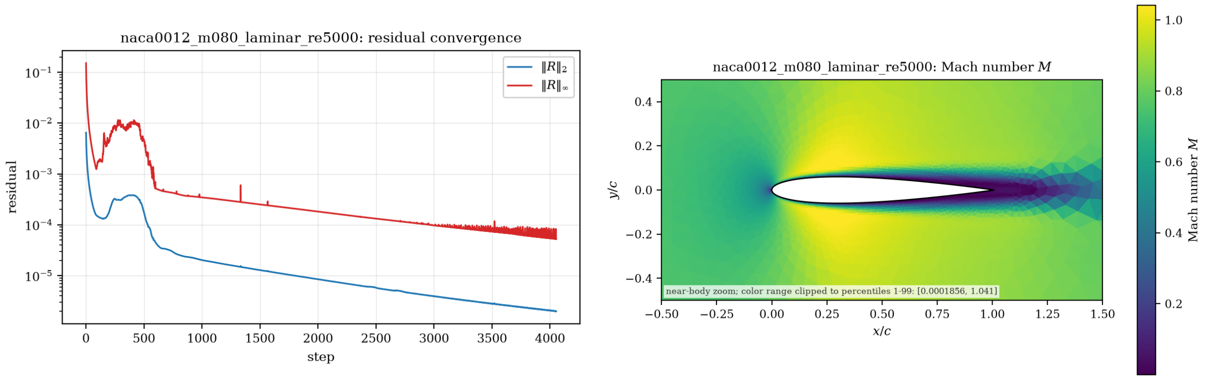

Figure 9: Mach 0.8, $Re = 5000$: residual history (left) and Mach contours (right).

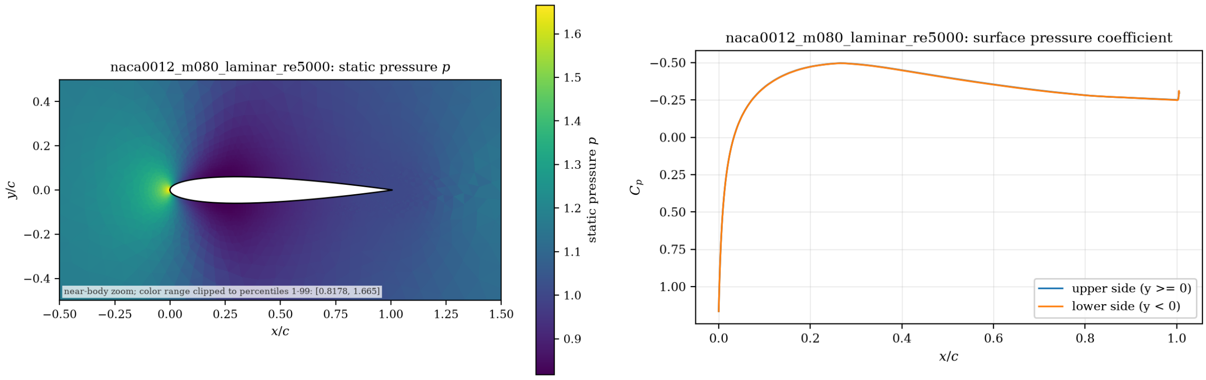

Figure 10: Mach 0.8, $Re = 5000$: pressure contours (left) and c_p distribution (right).

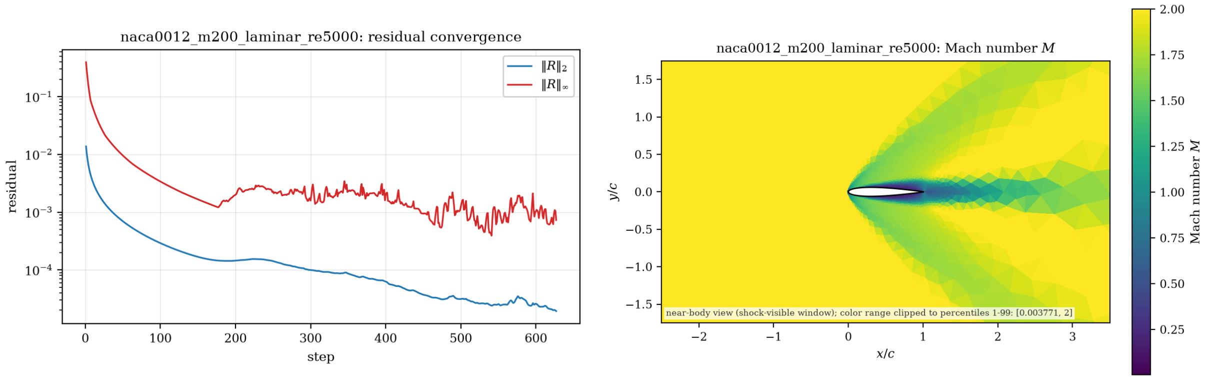

Figure 11: Mach 2.0, $Re = 5000$: residual history (left) and Mach contours (right).

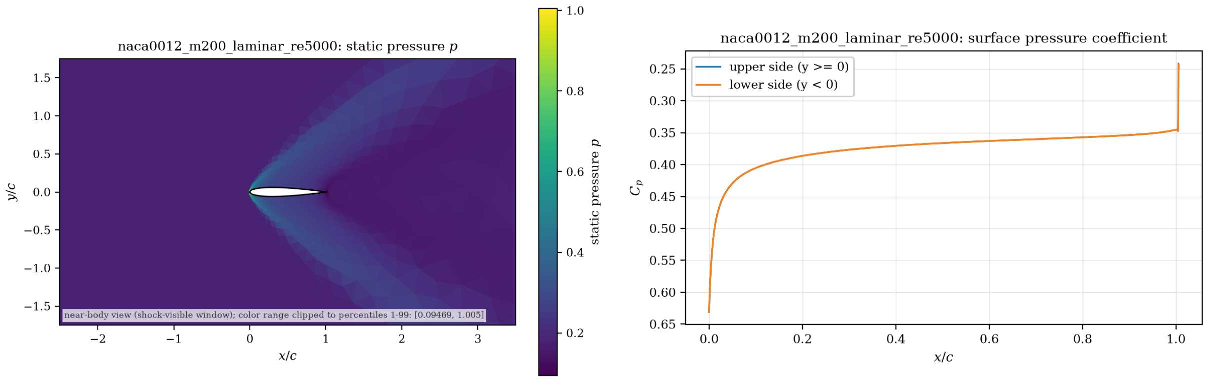

Figure 12: Mach 2.0, $Re = 5000$: pressure contours (left) and c_p distribution (right).

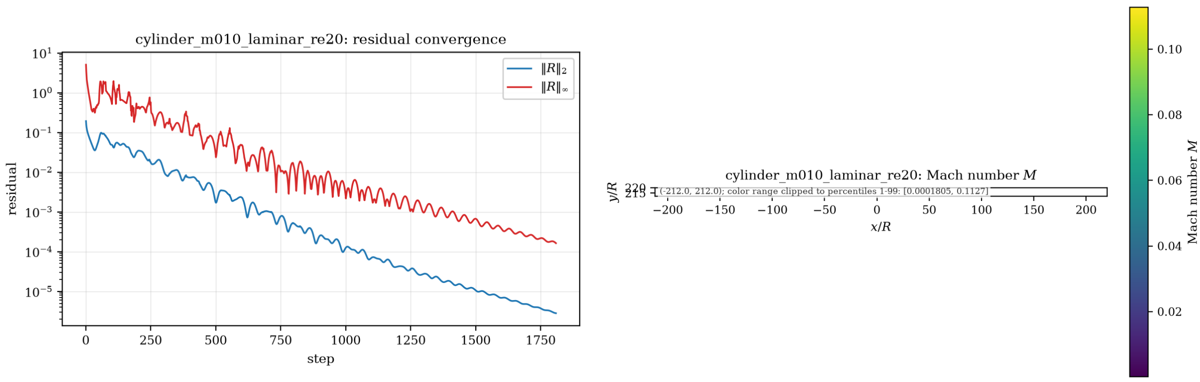

Figure 13: Cylinder $Re = 20$: residual history (left) and near-wake Mach contours (right).

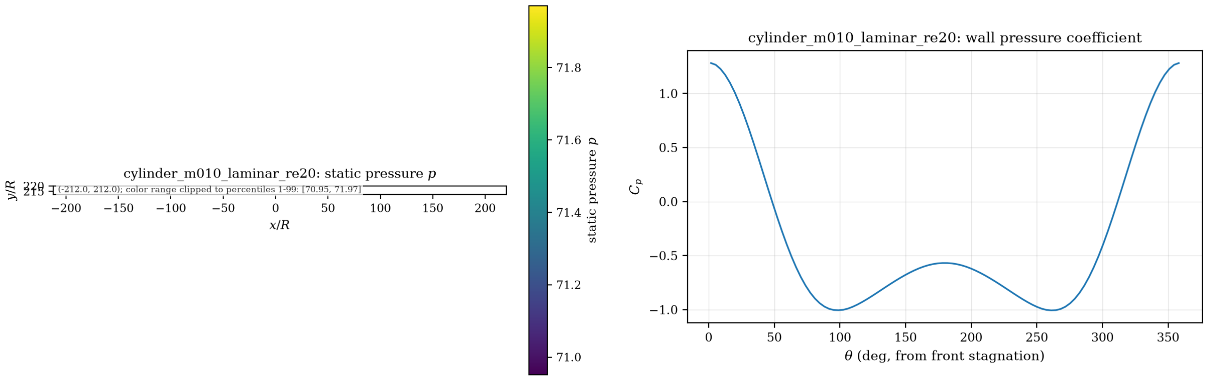

Figure 14: Cylinder $Re = 20$: pressure contours (left) and wall c_p versus angle (right).

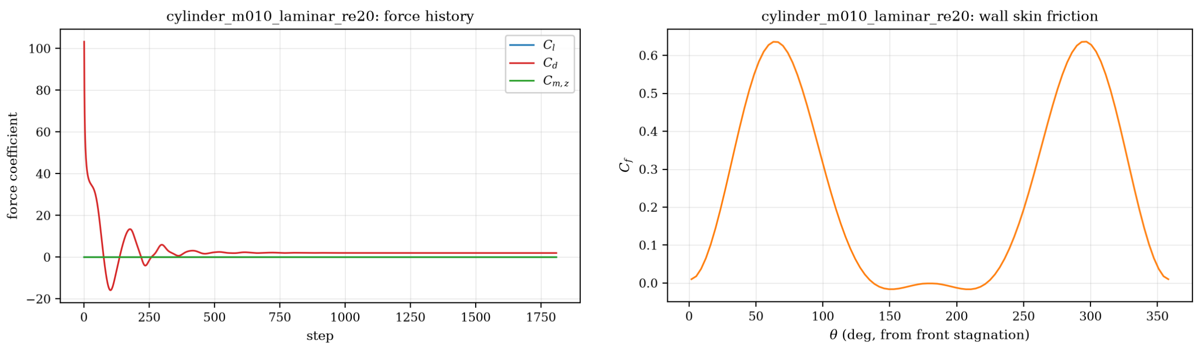

Figure 15: Cylinder $Re = 20$: force history (left) and wall skin-friction distribution (right).

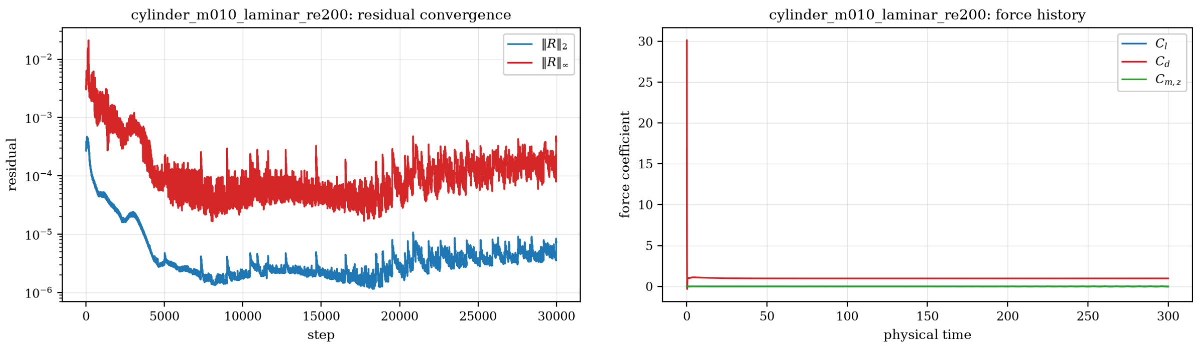

Figure 16: Cylinder $Re = 200$: residual history (left) and force history sampled by physical time (right).

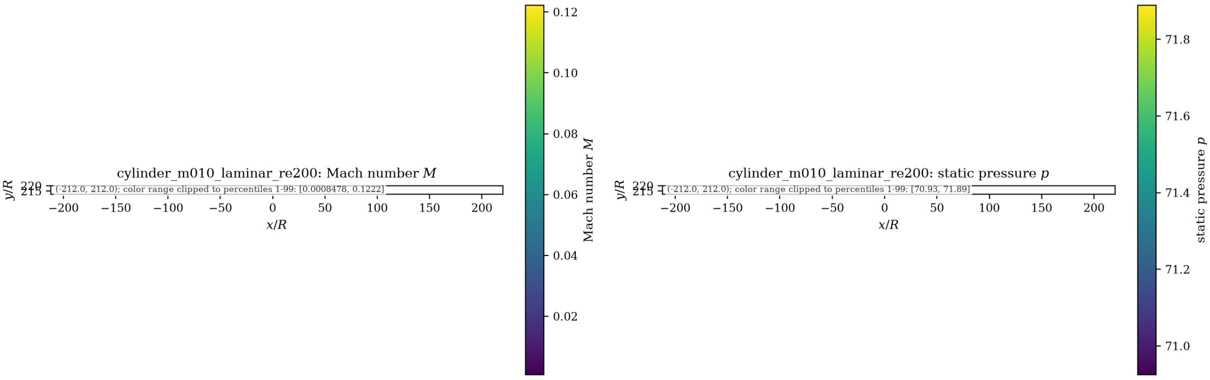

Figure 17: Cylinder $Re = 200$: Mach (left) and pressure (right) fields.

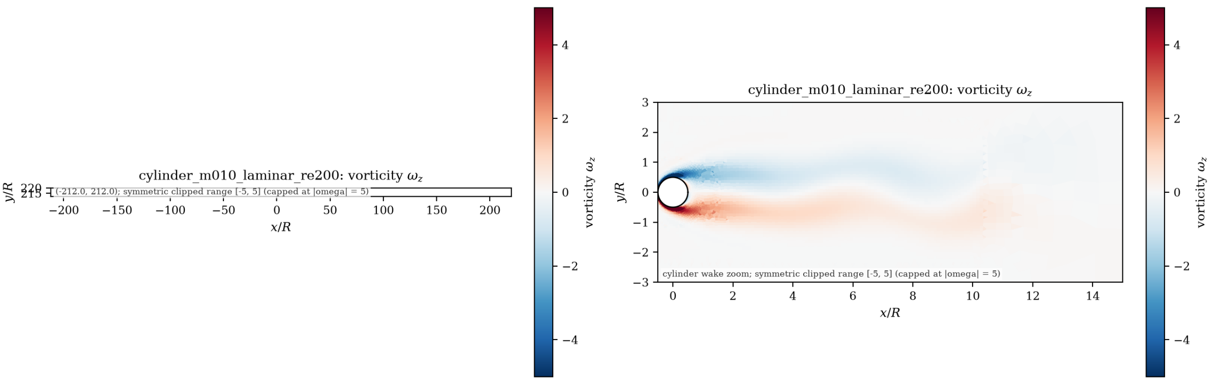

Figure 18: Cylinder $Re = 200$: vorticity (full view, left; wake zoom, right), clip range $[-5, 5]$.